QUESTION-ANSWER:- FOR LOOP

Q1. What is a for loop in Python?

✔Answer:

A **for loop** in Python is used to repeat a block of code for each item in a sequence (like list, string, or range). It helps in automating repetitive tasks.

Example use: printing numbers, iterating lists, etc.

Q2. Difference between for loop and while loop

✔Answer:

For Loop	While Loop
Used when number of iterations is known	Used when condition is unknown
Works on sequences (list, range, string)	Works on conditions
Easier to use	Slightly complex

Q3. Why is indentation important?

✔Answer:

Indentation defines the **block of code inside the loop**.

If indentation is wrong:

- Python gives an error OR
- Code runs incorrectly

Q4. What can we iterate using a for loop?

✔Answer:

We can iterate:

- Lists
- Strings
- Tuples
- Sets
- Dictionaries
- Ranges

Q5. Print numbers from 1 to 10

Code:

```
for i in range(1, 11):
    print(i)
```

Output:

```
1
2
3
4
5
6
7
8
9
10
```

Q6. Sum of first 10 natural numbers

Code:

```
sum = 0

for i in range(1, 11):
    sum += i

print("Sum =", sum)
```

Output:

Sum = 55

Q7. Print even numbers between 1 and 20

Code:

```
for i in range(1, 21):
    if i % 2 == 0:
        print(i)
```

Output:

2
4
6
8
10
12
14
16
18
20

Q8. Multiplication table

Code:

```
n = int(input("Enter number: "))

for i in range(1, 11):
    print(n, "x", i, "=", n * i)
```

Sample Input:

5

Output:

```
5 x 1 = 5
5 x 2 = 10
5 x 3 = 15
5 x 4 = 20
5 x 5 = 25
5 x 6 = 30
5 x 7 = 35
5 x 8 = 40
5 x 9 = 45
5 x 10 = 50
```

Q9. Count vowels in a string

Code:

```
text = input("Enter string: ")
vowels = "aeiouAEIOU"
count = 0

for char in text:
    if char in vowels:
        count += 1

print("Vowel count =", count)
```

Q10: How does a for loop work with a string in Python?

✔Answer:

A **for loop** can be used to iterate through a string character by character. In this case, each character of the string is accessed one by one and processed inside the loop.

This is useful when we want to perform operations like counting characters, searching letters, or modifying text.

Input:

Hello World

Output:

Vowel count = 3